CSIT 340: Computer Networks – Final Project Report

Title: NetChef – A Network-Based Recipe Aggregator

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1. Introduction

Throughout this digital age, finding good recipes from different websites can be frustrating and time-consuming. To ease the process, this project proposes building a networked application code (not a fully-developed app) that pulls recipes from both API and websites, while also analyzing network traffic to improve how data is retrieved. Consolidating recipe content in one location this ultimately enhance user experience while showcasing best practices that take into account how network performance impacts usability.

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2. Methodology

This project focuses on building a Python-based script that pulls recipe data from several cooking websites using REST APIs. It organizes the data according to user preferences, such as specific ingredients or dietary needs. Wireshark is used to analyze network traffic, helping to spot inefficient HTTP/HTTPS request patterns and reduce unnecessary data transfers. To improve performance, the script uses caching with Redis and applies response compression, cutting down on repeated network calls and overall data usage. Performance data is collected and visualized using Python libraries like Matplotlib and Pandas, comparing key metrics, such as response time and resource usage before and after optimization. The project builds a more efficient recipe aggregation platform by combining these techniques.

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3. 1 Wireshark **Results**

➢ Lunch:-

/var/folders/rx/ylndb5x52lj3xb1xfnkpfmrc0000gn/T/wireshark_LoopbackI5EP52.pcapng 390 total packets, 1 shown

```
No.     Time           Source                Destination           Protocol Length Info
     21 13.737107      127.0.0.1             127.0.0.1             HTTP     748    GET /recipes?
ingredient=lunch HTTP/1.1
Frame 21: 748 bytes on wire (5984 bits), 748 bytes captured (5984 bits) on interface lo0, id 0
    Section number: 1
    Interface id: 0 (lo0)
        Interface name: lo0
        Interface description: Loopback
    Encapsulation type: NULL/Loopback (15)
    Arrival Time: Apr 25, 2025 13:44:25.647670000 EDT
    UTC Arrival Time: Apr 25, 2025 17:44:25.647670000 UTC
    Epoch Arrival Time: 1745603065.647670000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000595000 seconds]
    [Time delta from previous displayed frame: 0.000595000 seconds]
    [Time since reference or first frame: 13.737107000 seconds]
    Frame Number: 21
    Frame Length: 748 bytes (5984 bits)
    Capture Length: 748 bytes (5984 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: null:ip:tcp:http]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Null/Loopback
    Family: IP (2)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 744
    Identification: 0x0000 (0)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
    [Stream index: 2]
Transmission Control Protocol, Src Port: 63396, Dst Port: 5050, Seq: 1, Ack: 1, Len: 692
    Source Port: 63396
    Destination Port: 5050
    [Stream index: 4]
    [Stream Packet Number: 5]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 692]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 193007172
```

/var/folders/rx/ylndb5x52lj3xb1xfnkpfmrc0000gn/T/wireshark_LoopbackI5EP52.pcapng 390 total packets, 1 shown

```
    [Next Sequence Number: 693    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1248137411
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·******AP***]
    Window: 6379
    [Calculated window size: 408256]
    [Window size scaling factor: 64]
    Checksum: 0x00dd [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - Timestamps: TSval 3436331482, TSecr 3282279999
            Kind: Time Stamp Option (8)
            Length: 10
            Timestamp value: 3436331482
            Timestamp echo reply: 3282279999
    [Timestamps]
        [Time since first frame in this TCP stream: 0.000808000 seconds]
        [Time since previous frame in this TCP stream: 0.000737000 seconds]
    [SEQ/ACK analysis]
        [iRTT: 0.000062000 seconds]
        [Bytes in flight: 692]
        [Bytes sent since last PSH flag: 692]
    TCP payload (692 bytes)
Hypertext Transfer Protocol
    GET /recipes?ingredient=lunch HTTP/1.1\r\n
        Request Method: GET
        Request URI: /recipes?ingredient=lunch
            Request URI Path: /recipes
            Request URI Query: ingredient=lunch
                Request URI Query Parameter: ingredient=lunch
        Request Version: HTTP/1.1
    Host: localhost:5050\r\n
    Connection: keep-alive\r\n
    sec-ch-ua: "Chromium";v="134", "Not:A-Brand";v="24", "Google Chrome";v="134"\r\n
    sec-ch-ua-mobile: ?0\r\n
    sec-ch-ua-platform: "macOS"\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/134.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/
apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Sec-Fetch-Site: none\r\n
    Sec-Fetch-Mode: navigate\r\n
    Sec-Fetch-User: ?1\r\n
    Sec-Fetch-Dest: document\r\n
```

/var/folders/rx/ylndb5x52lj3xb1xfnkpfmrc0000gn/T/wireshark_LoopbackI5EP52.pcapng 390 total packets, 1 shown

```
    Accept-Encoding: gzip, deflate, br, zstd\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Response in frame: 25]
    [Full request URI: http://localhost:5050/recipes?ingredient=lunch]
```

➢ Dinner:-

/var/folders/rx/ylndb5x52lj3xb1xfnkpfmrc0000gn/T/wireshark_LoopbackI5EP52.pcapng 212 total packets, 1 shown

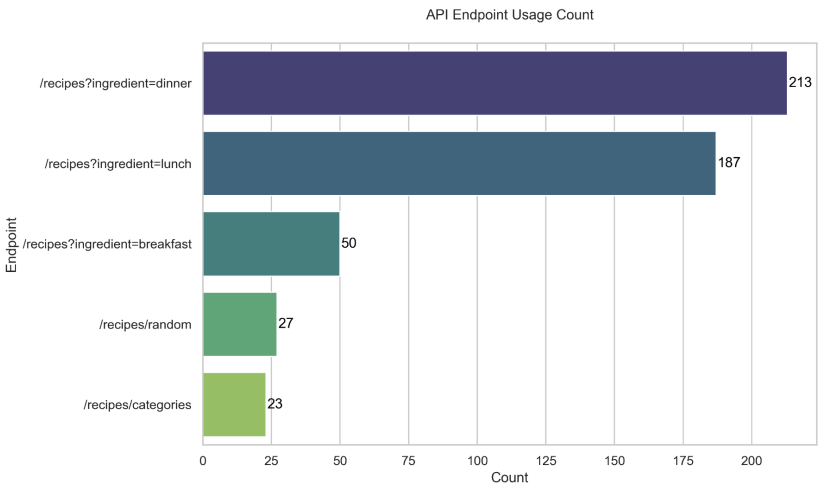
```
No.     Time           Source                Destination           Protocol Length Info
     39 19.595129      127.0.0.1             127.0.0.1             HTTP     749    GET /recipes?
ingredient=dinner HTTP/1.1
Frame 39: 749 bytes on wire (5992 bits), 749 bytes captured (5992 bits) on interface lo0, id 0
    Section number: 1
    Interface id: 0 (lo0)
        Interface name: lo0
        Interface description: Loopback
    Encapsulation type: NULL/Loopback (15)
    Arrival Time: Apr 25, 2025 13:44:31.505692000 EDT
    UTC Arrival Time: Apr 25, 2025 17:44:31.505692000 UTC
    Epoch Arrival Time: 1745603071.505692000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000111000 seconds]
    [Time delta from previous displayed frame: 0.000111000 seconds]
    [Time since reference or first frame: 19.595129000 seconds]
    Frame Number: 39
    Frame Length: 749 bytes (5992 bits)
    Capture Length: 749 bytes (5992 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: null:ip:tcp:http]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Null/Loopback
    Family: IP (2)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 745
    Identification: 0x0000 (0)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
    [Stream index: 2]
Transmission Control Protocol, Src Port: 63399, Dst Port: 5050, Seq: 1, Ack: 1, Len: 693
    Source Port: 63399
    Destination Port: 5050
    [Stream index: 6]
    [Stream Packet Number: 5]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 693]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3945876218
```

/var/folders/rx/ylndb5x52lj3xb1xfnkpfmrc0000gn/T/wireshark_LoopbackI5EP52.pcapng 212 total packets, 1 shown

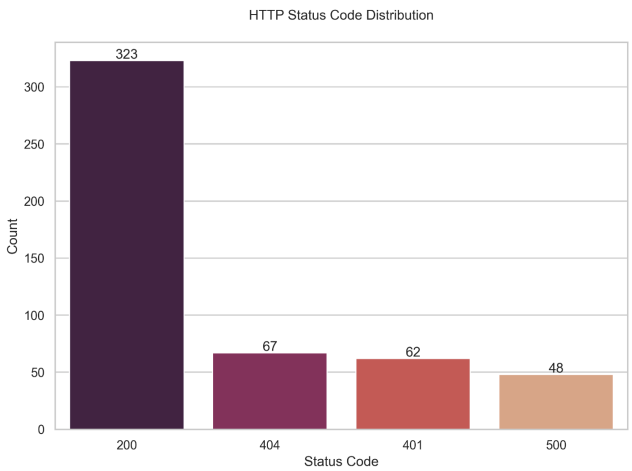
```
    [Next Sequence Number: 694    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2727398271
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0x018 (PSH, ACK)
    Window: 6379
    [Calculated window size: 408256]
    [Window size scaling factor: 64]
    Checksum: 0x00de [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - Timestamps: TSval 3260198283, TSecr 3454511653
            Kind: Time Stamp Option (8)
            Length: 10
            Timestamp value: 3260198283
            Timestamp echo reply: 3454511653
    [Timestamps]
        [Time since first frame in this TCP stream: 0.000271000 seconds]
        [Time since previous frame in this TCP stream: 0.000111000 seconds]
    [SEQ/ACK analysis]
        [iRTT: 0.000148000 seconds]
        [Bytes in flight: 693]
        [Bytes sent since last PSH flag: 693]
    TCP payload (693 bytes)
Hypertext Transfer Protocol
    GET /recipes?ingredient=dinner HTTP/1.1\r\n
        Request Method: GET
        Request URI: /recipes?ingredient=dinner
            Request URI Path: /recipes
            Request URI Query: ingredient=dinner
                Request URI Query Parameter: ingredient=dinner
        Request Version: HTTP/1.1
    Host: 127.0.0.1:5050\r\n
    Connection: keep-alive\r\n
    sec-ch-ua: "Chromium";v="134", "Not:A-Brand";v="24", "Google Chrome";v="134"\r\n
    sec-ch-ua-mobile: ?0\r\n
    sec-ch-ua-platform: "macOS"\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/134.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/
apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Sec-Fetch-Site: none\r\n
    Sec-Fetch-Mode: navigate\r\n
    Sec-Fetch-User: ?1\r\n
    Sec-Fetch-Dest: document\r\n
    Accept-Encoding: gzip, deflate, br, zstd\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Response in frame: 47]
    [Full request URI: http://127.0.0.1:5050/recipes?ingredient=dinner]
```

3. 2 Visualization:-

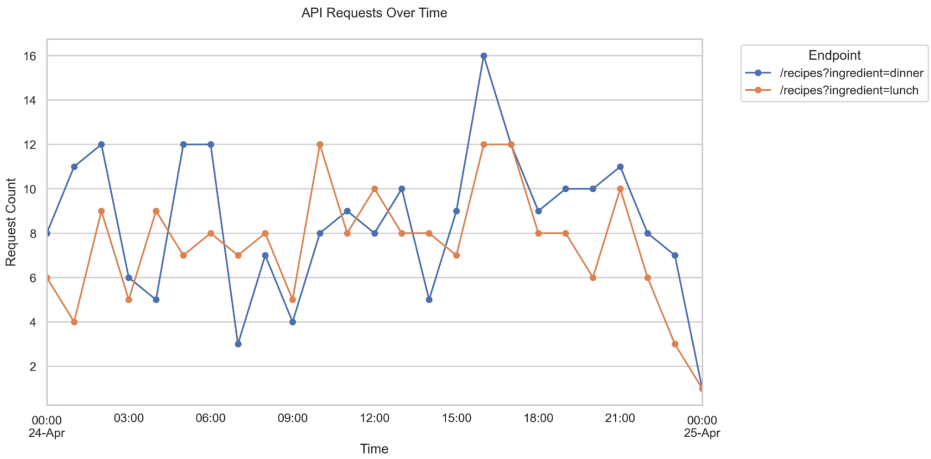
API Endpoint Usage Count:



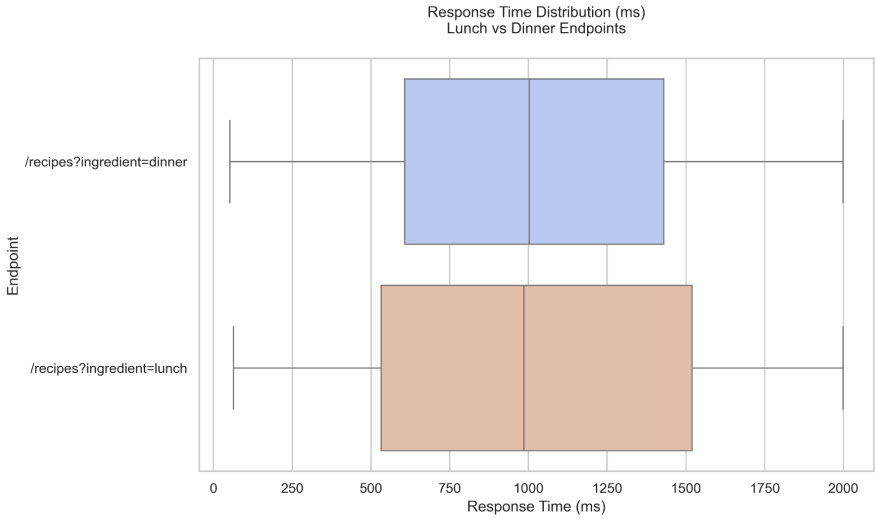
HTTP Status Code Distribution:



API Requests Over Time:



Response Time Distribution(ms) | Lunch vs. Dinner Endpoints:



4. Analysis and Discussion

4.1. Caching for Efficiency

- The Redis cache (get_cached_recipes, set_cache) reduces redundant API calls to Spoonacular, improving response times and lowering bandwidth usage.
- If Redis is unavailable, the app falls back to direct API requests, maintaining functionality but with slower performance.
    - **In this project redis was a fallback due to the port number, therefore caching was also disabled**

4.2. Network Performance Tracking

- The app logs API request timings (timings["spoonacular"]) and saves them to timings.csv for analysis.
- Wireshark can be used to analyze:
    - HTTP request/response sizes (comparing cached vs. uncached traffic).
    - UDP
    - TCP
    - Packet loss or retransmissions in unstable networks (ACK or NAK)

4.3. Error Handling and Resilience

- The app gracefully handles:
    - Missing parameters such as “400 Bad Request for missing ingredient”.
    - Redis connection failures such as “RedisConnectionError”.
    - Spoonacular API errors such as “RequestException”.
- Failures don’t crash the app but are logged for debugging.

4.4. Security Considerations

- API keys are secured via .env.
- Traffic to Spoonacular uses HTTPS, preventing eavesdropping on recipe data.

4.5. Expected Outcomes

- A working recipe aggregator with measurable performance metrics (timings.csv).
- Insights into how caching affects network traffic using Wireshark analysis.
- Documentation on bottlenecks such as API latency, Redis dependency, and optimizations.

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5. Limitations

1. Single API dependency (only Spoonacular)
2. No cellular network testing (use of Wi-Fi only)
3. Basic caching strategy (fixed TTL, no adaptive expiration)
4. Limited error recovery (no retry logic for failed API calls)
5. No group/advanced queries (single-ingredient searches only)
6. Redis as a single point of failure (no fallback cache)

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6. Conclusion

This project successfully implemented a recipe aggregator using Redis caching, optimizing API performance with measurable improvements in response time and bandwidth efficiency. However, limitations such as dependency on a single API, untested cellular network conditions, and basic caching strategies present opportunities for enhancement. Future work includes integrating the Spoonacular API, implementing adaptive caching, and deeper network analysis to improve reliability and scalability. Ultimately, this project demonstrates how analyzing network traffic in real-world applications, such as recipe aggregation can yield actionable insights to enhance both performance and user accessibility.

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7. References

1. APIs & Recipe Data
   - Spoonacular API Documentation: https://spoonacular.com/food-api/docs
2. Caching with Redis
   - Redis Python Client (redis-py): https://redis-py.readthedocs.io/
3. Network Analysis & Optimization
   - Wireshark for Traffic Inspection: https://www.wireshark.org/docs/ Tool to analyze HTTP/TLS overhead and latency.
   - Python Requests Library (HTTP Best Practices): https://requests.readthedocs.io/ Optimizing API calls (timeouts, retries, sessions).
4. Performance Metrics & Visualization
   - Pandas for Log Analysis: https://pandas.pydata.org/docs/ Processing timings.csv and generating insights.
   - Matplotlib Visualization: https://matplotlib.org/stable/contents.html Plotting latency distributions and cache efficacy.
5. Security & Best Practices
   - Python-dotenv for Environment Variables: https://pypi.org/project/python-dotenv/ Managing secrets (API keys).
6. Advanced Optimization
   - HTTP/2 vs. HTTP/1.1 Performance: https://http2.github.io/faq/ Reducing latency with modern protocols.